

Arbitrary c_0 -Sums Preserve Weak Stability Exactly

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Status. Candidate substantial partial result for arXiv:2012.04940. Likely valid; human review requested.

1 Source question

A unit ball is *weakly stable* if every finite convex combination of relatively weakly open subsets of the ball is again relatively weakly open. This is property CWO in the terminology of Abrahamsen et al. [2].

López-Pérez and Medina prove that $B_{C_0(K,X)}$ is weakly stable when K is scattered and B_X is both norm stable and weakly stable. Conversely, weak stability of $B_{C_0(K,X)}$ forces weak stability of both $B_{C_0(K)}$ and B_X ; the first condition forces K to be scattered. They then ask whether these necessary conditions are sufficient without the norm-stability hypothesis [1, p. 13].

Proposition 3.7 turns Theorem 2.5 (b) of [2] into a complete characterization since weak and norm stability agree on finite dimensional spaces. Observe that the weak stability of the unit ball in $C_0(K, X)$ implies the weak stability of the unit ball in both $C_0(K)$ and X [2], since $C_0(K)$ and X are 1-complemented in $C_0(K, X)$. We don't know if the converse is true.

We prove the full converse for every discrete K , and in fact for arbitrary c_0 -sums with varying fibers.

2 Claimed contribution

For a family $(X_\gamma)_{\gamma \in \Gamma}$, write

$$Z = \left(\bigoplus_{\gamma \in \Gamma} X_\gamma \right)_{c_0} = \left\{ z = (z_\gamma) : \sup_\gamma \|z_\gamma\| < \infty, \{ \gamma : \|z_\gamma\| \geq \varepsilon \} \text{ finite for all } \varepsilon > 0 \right\}.$$

Theorem 1. *Let Γ be nonempty and $(X_\gamma)_{\gamma \in \Gamma}$ be arbitrary real or complex Banach spaces. Then B_Z is weakly stable if and only if every B_{X_γ} is weakly stable.*

Corollary 1. *If K is any discrete locally compact space and X any Banach space, then*

$$B_{C_0(K,X)} \text{ is weakly stable} \iff B_X \text{ is weakly stable.}$$

Equivalently, for every nonempty set Γ and every weakly stable X ,

$$c_0(\Gamma) \widehat{\otimes}_\varepsilon X \cong c_0(\Gamma, X)$$

has a weakly stable unit ball.

3 Idea of the proof

The dual of a c_0 -sum is an ℓ_1 -sum. Hence finitely many weak tests can be approximated simultaneously by tests supported on one finite coordinate set F . Shrink each original weak neighborhood so that controlling weak coordinates on F is sufficient to remain inside it.

Now split a nearby target independently in every coordinate of F , using weak stability of the corresponding fiber. No continuous selection is required: F is finite and discrete. Outside F , put the target coordinate itself in every component of the convex combination. The resulting component vectors still vanish at infinity, because they differ from the target at only finitely many coordinates. This gives the desired weak neighborhood of the convex combination.

4 Proof

We first isolate the finite-coordinate reduction.

Lemma 1. *Let $z \in B_Z$ and let U be a relatively weakly open neighborhood of z in B_Z . There are a finite set $F \subset \Gamma$ and, for every $\gamma \in F$, a relatively weakly open neighborhood U_γ of z_γ in B_{X_γ} such that*

$$w \in B_Z, \quad w_\gamma \in U_\gamma \ (\gamma \in F) \implies w \in U. \quad (1)$$

Proof. Since

$$Z^* = \left(\bigoplus_{\gamma \in \Gamma} X_\gamma^* \right)_{\ell_1},$$

choose $f_1, \dots, f_r \in Z^*$ and $\eta > 0$ such that the basic neighborhood

$$V = \{w \in B_Z : |f_q(w - z)| < \eta, \ 1 \leq q \leq r\}$$

is contained in U . Approximate every f_q in dual norm by a finitely supported g_q so that $2\|f_q - g_q\| < \eta/2$, and let F contain all their supports. If F is empty there is nothing to prove. Otherwise, writing $g_q = (g_{q,\gamma})_{\gamma \in F}$, put

$$U_\gamma = \left\{ u \in B_{X_\gamma} : |g_{q,\gamma}(u - z_\gamma)| < \frac{\eta}{2|F|} \text{ for } 1 \leq q \leq r \right\}.$$

If w satisfies the left side of (1), then

$$|g_q(w - z)| \leq \sum_{\gamma \in F} |g_{q,\gamma}(w_\gamma - z_\gamma)| < \eta/2,$$

while $|(f_q - g_q)(w - z)| \leq 2\|f_q - g_q\| < \eta/2$. Thus $w \in V \subset U$. □

Assume first that every B_{X_γ} is weakly stable. Let $U_1, \dots, U_m$ be relatively weakly open subsets of B_Z , let $\lambda_j > 0$ with $\sum_j \lambda_j = 1$, and fix

$$z = \sum_{j=1}^m \lambda_j z^j, \quad z^j \in U_j.$$

Apply Lemma 1 to every pair (z^j, U_j) and enlarge the finitely many resulting supports to a common finite set F . Taking $U_{j,\gamma} = B_{X_\gamma}$ on any newly added coordinate, we obtain weak neighborhoods $U_{j,\gamma}$ of z^j such that coordinate membership on F forces membership in U_j .

For $\gamma \in F$, weak stability of B_{X_γ} makes

$$W_\gamma = \sum_{j=1}^m \lambda_j U_{j,\gamma}$$

relatively weakly open in B_{X_γ} and containing z_γ . Therefore

$$W = \{w \in B_Z : w_\gamma \in W_\gamma \text{ for every } \gamma \in F\}$$

is a relatively weakly open neighborhood of z in B_Z , because the finite coordinate projections are weak-to-weak continuous.

Take $w \in W$. For every $\gamma \in F$, choose $u_\gamma^j \in U_{j,\gamma}$ with $w_\gamma = \sum_j \lambda_j u_\gamma^j$. Define

$$w_\gamma^j = \begin{cases} u_\gamma^j, & \gamma \in F, \\ w_\gamma, & \gamma \notin F. \end{cases}$$

Each w^j belongs to B_Z : every coordinate has norm at most one, and w^j differs from the c_0 vector w on only finitely many coordinates. The finite-coordinate reduction gives $w^j \in U_j$, and $w = \sum_j \lambda_j w^j$. Hence $W \subset \sum_j \lambda_j U_j$, proving weak stability of B_Z .

Conversely, suppose B_Z is weakly stable and fix $\gamma_0 \in \Gamma$. Let $J : X_{\gamma_0} \rightarrow Z$ be the coordinate embedding and $P : Z \rightarrow X_{\gamma_0}$ the coordinate projection. Both have norm one and $PJ = I$. If $A_j \subset B_{X_{\gamma_0}}$ are relatively weakly open, then

$$\tilde{A}_j = \{z \in B_Z : Pz \in A_j\}$$

are relatively weakly open. Given $x = \sum_j \lambda_j a_j$ with $a_j \in A_j$, weak stability of B_Z says that Jx is an interior point of $\sum_j \lambda_j \tilde{A}_j$ in B_Z . Intersect a witnessing neighborhood with $J(B_{X_{\gamma_0}})$ and pull it back by J . If $Jy = \sum_j \lambda_j z^j$ with $z^j \in \tilde{A}_j$, applying P gives $y = \sum_j \lambda_j Pz^j$ with $Pz^j \in A_j$. Thus x is an interior point of $\sum_j \lambda_j A_j$. The coordinate unit ball is weakly stable, completing the proof of Theorem 1.

For Corollary 1, use $C_0(K, X) = c_0(K, X)$ when K is discrete. On finite tensors $\sum_{\gamma \in F} e_\gamma \otimes x_\gamma$, the injective tensor norm is $\max_{\gamma \in F} \|x_\gamma\|$ (one inequality is immediate, and the other uses a coordinate functional and a norming functional for x_γ). Completion yields the stated isometry.

5 Scope, comparison, and novelty

Theorem 5.2 of Abrahamsen et al. proves a predecessor: $c_0(X_n)$ has CWO when the fibers are finite-dimensional and satisfy the stronger norm-selection property (co) [2, Theorem 5.2]. Their

proof uses finite-dimensionality to turn weak coordinate control into norm control and then invokes continuous norm selections. Lemma 1 keeps the coordinate conditions weak, so the present proof removes finite-dimensionality and (co), permits arbitrary fibers, and permits an arbitrary index set.

This does not settle the source question for a non-discrete scattered K . At a non-isolated point, pointwise decompositions must assemble into norm-continuous X -valued functions. The source obtains that assembly from norm stability and Michael selection. Already for the one-point compactification of a sequence, the missing step is the norm lower semicontinuity of the inverse midpoint map, closely tied to the separate open question whether weak stability implies norm stability.

A bounded novelty search on 11 August 2026 covered the four lightweight run indexes and the two source papers. Exact searches combined arbitrary c_0 -sums with weak stability and property CWO; further searches used the vector-valued converse and injective-tensor question. The search found the cited finite-dimensional (co) theorem, but no arbitrary weakly stable-coordinate version. This supports, but cannot certify, novelty.

Human-review recommendation. Review as a substantial partial solution. The key audit is the finite-support reduction; the remainder is an exact coordinate patching argument.

References

- [1] G. López-Pérez and R. Medina, *A characterization of the weak topology in the unit ball of purely atomic L_1 preduals*, J. Math. Anal. Appl. **504** (2021), 125434; arXiv:2012.04940.
- [2] T. A. Abrahamsen, J. Becerra Guerrero, R. Haller, V. Lima, and M. Põldvere, *Banach spaces where convex combinations of relatively weakly open subsets of the unit ball are relatively weakly open*, Studia Math. **250** (2020), 297–320; arXiv:1806.10693.